Ekaterina Bremel

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Summary

Computer Science student (**4.0 GPA, Dean's List**) focused on shipping AI that actually works in production. **Built LLM integrations and ML classification models across fintech and health tech**, **delivered 85% classification accuracy, optimized pipelines by 35%, and achieved 23% error reduction.** Proficient in **Python and modern ML frameworks.** I think about accuracy not as a metric, but as a responsibility.

Technical Skills

Languages: Python, SQL

Data Engineering & Pipelines: Pandas, NumPy, ETL Pipelines, Data Validation, Data Preprocessing, Structured & Unstructured Data

AI & Machine Learning: LLM (Anthropic Claude), Scikit-learn, TensorFlow, PyTorch, Keras, Ensemble Methods (Random Forest, CatBoost, LightGBM), Hyperparameter Tuning, Cross-Validation, Regression, Classification, Clustering, CNNs (ResNet50)

NLP & AI Engineering: Prompt Engineering, TF-IDF, NLTK, Text Classification

Data Analysis & Visualization: EDA, **Feature Engineering**, Statistical Analysis, Matplotlib, Seaborn

Tools & Additional: GitHub, Jira, Jupyter Notebook

Education

St. Francis College — Expected Graduation: December 2026
Bachelor of Science in Computer Science — Brooklyn, NY

- **Dean's List: All Semesters, Varsity Tennis Team - GPA: 4.0/4.0**
- Courses: Machine Learning, Software Engineering, Data Structures & Algorithms, Database Management Systems

Yandex School of Data Analysis
Data Science Certificate - GPA: 4.0/4.0

- Completed comprehensive ML curriculum including regression, classification, **NLP**, and computer vision

Work Experience

HumanKind OS — Apr 2025 - Present
Machine Learning Engineer (Part-time) — New York, NY

- **Developed Python data pipeline collecting, cleaning, and normalizing wearable device data**, computing Life Battery health scores, and storing results for user health profiles
- **Built LLM integration using Anthropic Claude**, engineering structured prompts with user health context to deliver personalized real-time AI responses
- **Contributed to designing Life Battery scoring algorithm**, researching and selecting key wearable metrics, defining metric weights, and validating scoring logic to ensure actionable results
- **Collaborated with cross-functional teams** (founders, engineering, marketing) to translate business requirements into technical solutions, shaping product strategy and go-to-market plans

Sberbank- Eastern Europe's largest bank, $500B+ assets, serves 108M+ clients — Feb 2024 - Apr 2025
Data Scientist

- **Increased customer feedback NLP classification accuracy to 85%** through data preprocessing, model selection, and hyperparameter tuning, optimizing the automated system used by product and marketing teams
- **Engineered scalable data preprocessing pipelines in Python and SQL, reducing processing time by 35%** through query-level tuning and workflow redesign, enabling large-scale feature engineering for predictive modeling
- **Improved predictive models on large-scale datasets using Python (Pandas, NumPy, Scikit-learn), achieving 23% error reduction** through feature engineering, model selection, and rigorous cross-validation experimentation, enabling finance and analytics teams to make data-driven decisions
- **Collaborated across engineering, analytics, product, and marketing teams**, participated in code reviews, translated model outputs into actionable business insights, and presented technical findings to non-technical stakeholders

Projects

Steel Temperature Prediction for Energy Optimization | Source Code — *Python | Scikit-learn| Machine Learning | CatBoost*

- **Developed predictive regression model** forecasting steel temperatures to optimize energy usage, achieving MAE of 4.05°C and **enabling 15-20% reduction in electricity consumption for manufacturing operations**

Toxic Comment Classification for Content Moderation | Source Code — *Python | NLTK |Matplotlib | Logistic Regression*

- **Built NLP classifier** for automated content moderation using TF-IDF, achieving F1 score of 0.78 (above 0.75 threshold), **reducing manual review workload by 60% through data-driven automation**

Customer Age Estimation using Computer Vision | Source Code — *TensorFlow | Computer Vision | ResNet50 | CNN*

- Created CNN using transfer learning to estimate customer age from photos, achieving MAE of 5.89 years, enabling targeted marketing campaigns and supporting age-restricted product compliance without manual verification

Gold Recovery Prediction for Ore Processing Optimization | Source Code — *Pandas | Random Forest | NumPy | Cross-Validation*

- **Built predictive model** for mining efficiency optimization, achieving sMAPE of 8.97**%, enabling operations teams to identify unprofitable batches and optimize resource allocation**